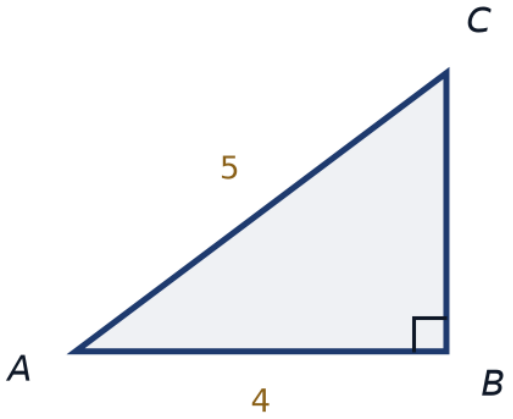
Advanced Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

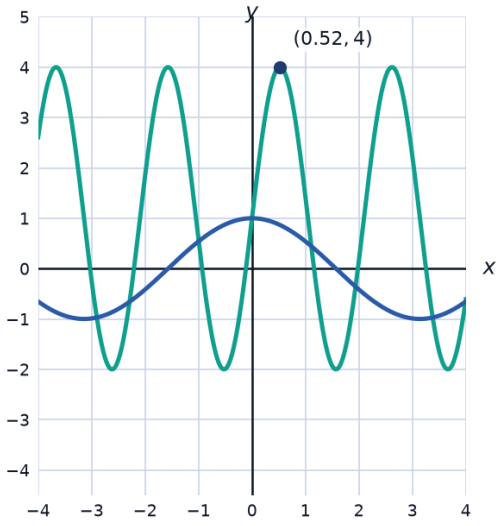
1 [Foundational] • Target: Trigonometric identities: Pythagorean and reciprocal

Given $\cos\theta = \frac{4}{5}$ and θ in the fourth quadrant, find $\sin\theta$ and $\tan\theta$.



2 [Foundational] • Target: Sum and difference identities

Use the sum identity to find the exact value of $\sin(75^\circ)$, using $\sin(75^\circ) = \sin(45^\circ + 30^\circ)$.



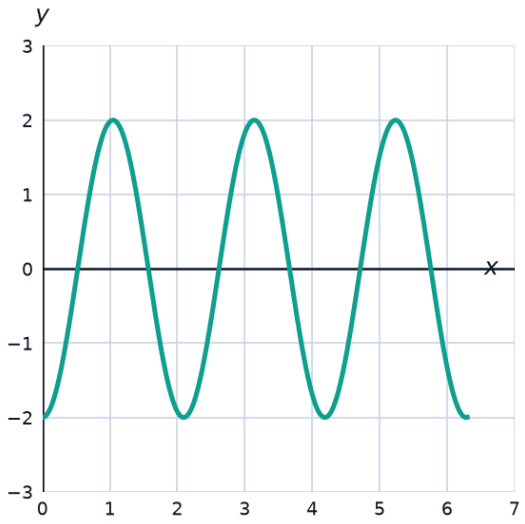
3 [Foundational] • Target: Double angle identities

Given $\sin\theta = \frac{3}{5}$ and θ in the first quadrant, find $\sin(2\theta)$ and $\cos(2\theta)$.

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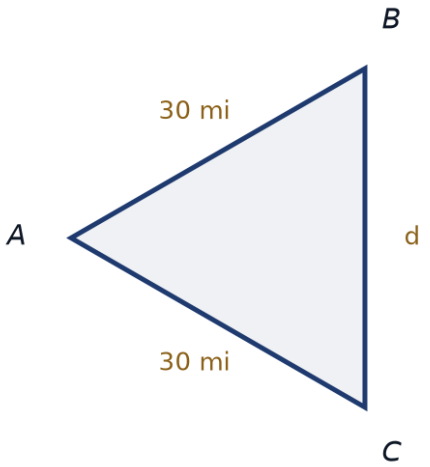
4 [Foundational] • Target: Graphs and transformations of trigonometric functions

State the amplitude, period, and phase shift of $y = 2\sin\left(3x - \frac{\pi}{2}\right)$.



5 [Foundational] • Target: Applications: Law of Sines and Law of Cosines together

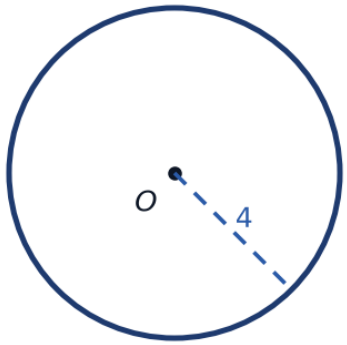
Two ships leave a port at the same time, one traveling at 20° and the other at 80° (measured from north), each at 15 mph. Find the distance between the ships after 2 hours, to the nearest mile, using the Law of Cosines.



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6 [Foundational] • Target: Polar coordinates

Convert the polar point $(4, \frac{\pi}{3})$ to rectangular coordinates.



7 [Foundational] • Target: A multi-part trigonometric identity verification problem

This question has two parts. (a) Use the identity $\tan\theta = \frac{\sin\theta}{\cos\theta}$ together with the double angle formulas to derive a formula for $\tan(2\theta)$ in terms of $\tan\theta$. (b) Use your formula to find $\tan(2\theta)$ given $\tan\theta = 3$.

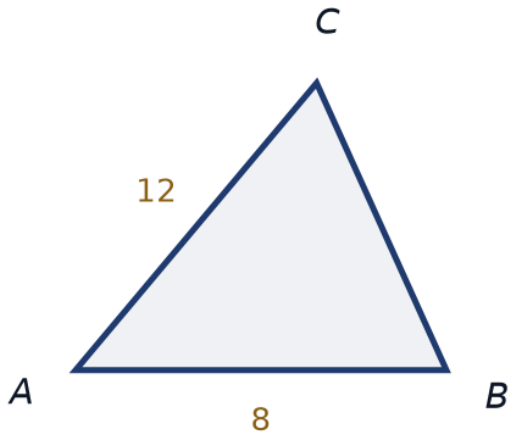
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Half-angle identities

Use the half-angle identity to find $\cos(15^\circ)$, given $\cos(30^\circ) = \frac{\sqrt{3}}{2}$.

9 [Intermediate] • Target: Area of a triangle using trigonometry

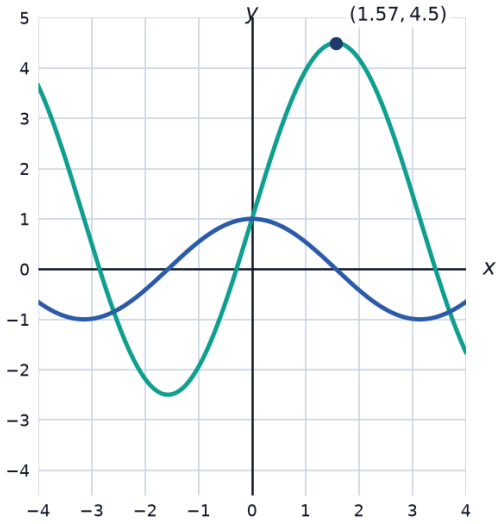
Find the area of a triangle with sides 8 and 12, and included angle 50°, using $\text{Area} = \frac{1}{2}ab\sin C$.



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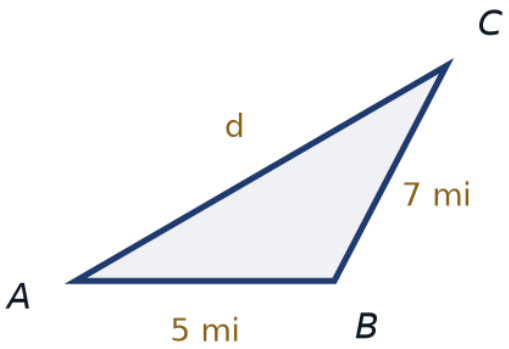
10 [Intermediate] • Target: Simplifying trigonometric expressions using identities

Simplify $\sin^2\theta + \cos^2\theta + \tan^2\theta$ in terms of $\sec\theta$.



11 [Intermediate] • Target: A multi-part navigation and triangulation problem

This question has three parts. A hiker walks 5 miles due east, then turns and walks 7 miles at a bearing of 40° north of east. (a) Find the angle between the two legs of the hike (measured as the interior angle at the turning point). (b) Use the Law of Cosines to find the straight-line distance from the start to the end point, to the nearest tenth of a mile. (c) Explain why this distance is less than the total distance walked (12 miles).



12 [Intermediate] • Target: Verifying a trigonometric identity

Verify the identity $\frac{1-\cos^2\theta}{\sin\theta} = \sin\theta$.

13 [Intermediate] • Target: Simplifying trigonometric expressions using identities

Prove the trigonometric identity $\frac{1-\cos 2\theta}{\sin 2\theta} = \tan\theta$ for all values of θ where both sides are defined.

14 [Intermediate] • Target: A multi-part navigation and triangulation problem

Find all solutions to $2\cos^2\theta - \cos\theta - 1 = 0$ on the interval $[0, 2\pi)$.

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Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Verifying a trigonometric identity • Cross-Lesson: 5.01 Pre-Calculus Concepts

In ΔABC, $a = 6$, $b = 10$, and angle $A = 30^\circ$. Determine how many possible triangles exist, and find the possible value(s) of angle B in degrees.

16 [Challenging] • Target: Trigonometric identities: Pythagorean and reciprocal • Cross-Lesson: 6.01 Advanced Coordinate Geometry

The sum of an arithmetic sequence is 150, with first term $a_1 = 5$ and last term $a_n = 25$. How many terms are in this sequence, and what is the common difference d?

17 [Challenging] • Target: Sum and difference identities • Cross-Lesson: 7.01 Advanced Plane Geometry

Calculate the exact value of $\cos(75^\circ)$ using angle addition, and evaluate $\tan(2\theta)$ given that $\tan\theta = \frac{2}{3}$.

18 [Challenging] • Target: Double angle identities • Cross-Lesson: 8.01 Statistics, Counting & Probability

In triangle ABC, $a = 10$, $b = 16$, and $A = 30^\circ$. Determine whether $0, 1,$ or 2 distinct triangles can be formed, and calculate all possible values for angle B to the nearest tenth of a degree.

19 [Challenging] • Target: Graphs and transformations of trigonometric functions • Cross-Lesson: 1.01 Advanced Number Theory

In triangle ABC, $a = 7$, $b = 5$, and $\cos C = \frac{1}{5}$. (a) Find the exact length of side c. (b) Calculate the exact value of $\sin^2 C$ and the area of triangle ABC.

20 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 2.01 Advanced Algebra & Systems

In triangle ABC, $a = 8$, $\angle B = 35^\circ$, and $\angle C = 75^\circ$. Use the Law of Sines to calculate the length of side b to two decimal places.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Trigonometric identities: Pythagorean and reciprocal

Given $\sin\theta = \frac{15}{17}$ and θ in the second quadrant, find $\cos\theta$.

a. $-\frac{15}{17}$

b. $\frac{15}{17}$

c. $-\frac{8}{17}$

d. $\frac{8}{17}$

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22 [Foundational] • Target: Sum and difference identities

Given $\cos\theta = -\frac{12}{13}$ and θ in the third quadrant, find $\sin\theta$.

a. $-\frac{12}{13}$

b. $\frac{12}{13}$

c. $-\frac{5}{13}$

d. $\frac{5}{13}$

23 [Foundational] • Target: Double angle identities

Two sides of a triangle measure 7 and 8, with an included angle of 120°. Find the length of the third side, using the Law of Cosines.

a. 14

b. 13

c. 12

d. 11

24 [Foundational] • Target: Graphs and transformations of trigonometric functions

In a triangle, sides $a = 15$ and $b = 14$ meet at an included angle of 123°. Using the Law of Cosines, which expression gives side c?

a. $\sqrt{15^2 + 14^2 + 2(15)(14)\cos 123^\circ}$

b. $\sqrt{15^2 - 14^2 - 2(15)(14)\cos 123^\circ}$

c. $\sqrt{15^2 + 14^2 - 2(15)(14)\cos 123^\circ}$

d. $15^2 + 14^2 - 2(15)(14)\cos 123^\circ$

25 [Foundational] • Target: Inverse trigonometric functions

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)

a. $6 - 24$

b. 22

c. $2 - 64$

d. $6 + 24$

26 [Foundational] • Target: Trigonometric equations with multiple solutions

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2 - 64$

b. $\frac{1}{2}$

c. 22

d. $6 - 24$

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27 [Foundational] • Target: Applications: Law of Sines and Law of Cosines together

In ΔABC, the side lengths opposite to angles A, B, and C are a, b, and c, respectively. If $\angle A - \angle B = 90^\circ$ and $a + b = \sqrt{2}c$, what is the measure of $\angle C$?

a. 30°

b. 45°

c. 60°

d. 75°

28 [Foundational] • Target: Polar coordinates

In trapezoid $ABCD$, base $\overline{AB}$ is parallel to base $\overline{CD}$, and $\angle DAB = \angle ADC = 90^\circ$. The length of $\overline{AB}$ is x, the length of $\overline{CD}$ is $2x + 1$, and the length of the non-parallel side $\overline{BC}$ is $2x$. If the height $AD = 8$, what is the area of trapezoid $ABCD$?

a. 34

b. 64

c. 80

d. 128

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: A multi-part trigonometric identity verification problem

In ΔABC, the length of side AB is x centimeters, the length of side BC is $(x + 3)$ centimeters, and the length of side AC is 7 centimeters. If the measure of $\angle B$ is 60°, what is the value of x?

a. 4

b. 5

c. 7

d. 8

30 [Intermediate] • Target: Half-angle identities

In the standard (x, y) coordinate plane, the terminal ray of an angle θ in standard position passes through the point $(-3, 4)$. What is the value of $\sec\theta$?

a. $-\frac{5}{3}$

b. $-\frac{3}{5}$

c. $\frac{5}{4}$

d. $-\frac{4}{3}$

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31 [Intermediate] • Target: Solving trigonometric equations that factor

In the figure below, ΔABC is a right triangle with right angle at B, and altitude $\overline{BD}$ is drawn perpendicular to hypotenuse $\overline{AC}$. If $AD = 4$ and $DC = 9$, what is the length of $\overline{BD}$?

a. $\sqrt{13}$

b. 6

c. 6.5

d. 13

32 [Intermediate] • Target: Area of a triangle using trigonometry

Two similar triangles have corresponding side lengths in ratio $4 : 6$. If the smaller triangle has area 8, what is the area of the larger triangle?

a. 10

b. 22

c. 14

d. 18

33 [Intermediate] • Target: Simplifying trigonometric expressions using identities

What are the equations of the asymptotes of the hyperbola $\frac{(x-1)^2}{9} - \frac{(y+2)^2}{4} = 1$?

a. $y + 2 = \pm\frac{2}{3}(x - 1)$

b. $y + 2 = \pm\frac{3}{2}(x - 1)$

c. $y = \pm\frac{2}{3}x$

d. $y - 2 = \pm\frac{2}{3}(x + 1)$

34 [Intermediate] • Target: A multi-part navigation and triangulation problem

An ellipse has equation $\frac{x^2}{36} + \frac{y^2}{20} = 1$. What are the coordinates of its foci?

a. $(\pm 5, 0)$

b. $(\pm 4, 0)$

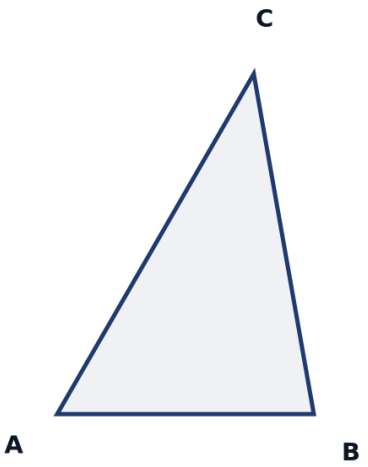
c. $(0, \pm 4)$

d. $(\pm 6, 0)$

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35 **[Intermediate]** • **Target: Verifying a trigonometric identity**

In ΔABC, $\angle A = 60^\circ$, $\angle B = 80^\circ$, and side $a = 5$. Find side b (to one decimal place) using the Law of Sines.



a. 5.7

b. 6.7

c. 7.7

d. 4.7

36 **[Intermediate]** • **Target: Trigonometric identities: Pythagorean and reciprocal**

Find the inverse of the matrix $M = (5, 2; 2, 1)$.

a. $(-1, 2; 2, -5)$

b. $(1, 2; 2, 5)$

c. $(5, -2; -2, 1)$

d. $(1, -2; -2, 5)$

37 **[Intermediate]** • **Target: Sum and difference identities**

Evaluate $(1 + i)^4$ using De Moivre's Theorem.

a. 4

b. -4

c. 16

d. -4i

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Double angle identities • Cross-Lesson: 8.01 Statistics, Counting & Probability

Evaluate the exact trigonometric value: $\sin\left(\frac{\pi}{6}\right)$.

a. $\frac{\sqrt{3}}{2}$

b. $\frac{\sqrt{2}}{2}$

c. 1

d. $\frac{1}{2}$

39 [Challenging] • Target: Graphs and transformations of trigonometric functions • Cross-Lesson: 1.01 Advanced Number Theory

Find the area of a triangle with base 12 and height 16.

a. 192

b. 96

c. 108

d. 80

40 [Challenging] • Target: Inverse trigonometric functions • Cross-Lesson: 2.01 Advanced Algebra & Systems

Prove the identity $\sin 3A = 3\sin A - 4\sin^3 A$ using the angle addition formula.

a. Not an identity

b. Only for $A < \pi/2$

c. Use calculus

d. $\sin(2A + A) = \sin 2A\cos A + \cos 2A\sin A = 2\sin A\cos^2 A + (1 - 2\sin^2 A)\sin A = 3\sin A - 4\sin^3 A$. ✓

41 [Challenging] • Target: Trigonometric equations with multiple solutions • Cross-Lesson: 3.01 Advanced Functions

In ΔABC, $a = 8$, $b = 10$, $c = 12$. Find $\angle A$ (to one decimal place) using the Law of Cosines.

a. 41.4°

b. 46.4°

c. 138.6°

d. 36.4°

42 [Challenging] • Target: Applications: Law of Sines and Law of Cosines together • Cross-Lesson: 5.01 Pre-Calculus Concepts

In ΔABC, $a = 5$, $b = 6$, $c = 7$. Find $\angle A$ (to one decimal place) using the Law of Cosines. (trial 2)

a. 49.4°

b. 135.6°

c. 44.4°

d. 39.4°

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43 [Challenging] • Target: Polar coordinates • Cross-Lesson: 6.01 Advanced Coordinate Geometry

Using a double-angle identity, evaluate $\cos^2\theta - \sin^2\theta$ when $\theta = 15^\circ$.

a. -1

b. $-\frac{\sqrt{3}}{2}$

c. $\frac{\sqrt{2}}{2}$

d. $\frac{\sqrt{3}}{2}$

44 [Challenging] • Target: A multi-part trigonometric identity verification problem • Cross-Lesson: 7.01 Advanced Plane Geometry

If $\cos\theta = \frac{1}{2}$ and $\frac{\theta}{2}$ is in Quadrant I, find $\sin\left(\frac{\theta}{2}\right)$ using the half-angle formula.

a. $\frac{1}{4}$

b. $\frac{1}{2}$

c. $\frac{3}{5}$

d. $-\frac{1}{2}$

45 [Challenging] • Target: Half-angle identities • Cross-Lesson: 8.01 Statistics, Counting & Probability

In ΔABC, $b = 10$, $c = 7$, and $\angle A = 120^\circ$. Find side a (to one decimal place) using the Law of Cosines.

a. 13.8

b. 14.8

c. 15.8

d. 12.2

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